

BATNA, Algeria

WMO: 604680

Lat: 35.752N

Lon: 6.309E

Elev: 822

StdP: 91.83

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	-1.9	-7.0	2.3	8.5	-5.5	2.6	6.0	14.4	10.2	12.6	9.6	1.3	120	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	39.2	18.8	37.8	18.5	36.1	18.3	21.2	32.5	20.4	31.5	19.7	30.9	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	14.4	24.5	17.1	13.5	23.8	16.2	12.7	23.1	65.5	32.7	62.2	31.4	59.6	30.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	10.7	9.4	-5.7	41.6	1.3	0.8	-6.6	42.2	-7.3	42.7	-8.0	43.1	-8.9	43.7		
			-6.1	23.0	1.5	1.6	-7.2	24.2	-8.1	25.1	-8.9	26.0	-9.9	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	6.2	6.9	10.3	13.5	18.3	23.4	27.2	26.4	21.7	17.2	10.9	7.2
	DBStd	8.04	2.55	3.11	3.51	3.58	4.04	3.89	2.92	2.78	3.37	3.43	3.20	2.74
	HDD10.0	383	120	95	39	8	0	0	0	0	0	1	26	93
	HDD18.3	1787	375	319	249	150	52	5	0	0	8	62	223	344
	CDD10.0	2513	3	8	49	113	257	403	533	507	352	225	53	8
	CDD18.3	876	0	0	1	4	50	158	275	249	110	28	1	0
	CDH23.3	12018	0	3	27	133	796	2229	3902	3324	1227	366	11	1
CDH26.7	6597	0	1	4	25	319	1212	2413	1968	541	113	1	0	
Wind	WSAvg	3.7	3.3	3.8	4.0	4.0	3.9	3.9	4.0	3.8	3.5	3.3	3.5	3.1
Precipitation	PrecAvg	367	40	31	37	40	42	16	7	20	35	28	32	39
	PrecMax	606	139	101	83	101	103	52	33	73	76	88	105	127
	PrecMin	226	8	0	4	4	3	0	0	2	10	7	9	0
	PrecStd	92	30	24	18	24	31	13	7	21	19	21	24	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.0	26.2	29.2	35.5	39.3	41.1	40.4	35.9	32.1	24.9	20.0
		MCWB	9.8	10.3	12.4	14.2	16.2	17.9	19.7	18.8	18.6	16.4	13.8	11.2
	2%	DB	16.2	18.8	23.1	27.0	32.2	37.2	39.9	38.8	33.9	29.8	21.9	17.0
		MCWB	9.1	9.7	11.3	13.5	16.1	17.6	19.4	18.7	18.1	15.6	12.2	10.1
	5%	DB	14.1	16.1	21.0	24.7	30.2	35.5	38.2	37.1	32.0	27.2	19.9	15.0
		MCWB	8.2	8.5	10.9	13.0	15.6	17.4	18.8	18.4	17.7	15.2	11.5	9.2
	10%	DB	12.2	14.0	18.8	22.1	28.1	33.2	36.8	35.4	30.0	24.9	17.9	13.2
		MCWB	7.3	7.7	10.1	12.3	15.2	17.1	18.6	18.5	17.7	14.8	10.9	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.6	13.7	16.5	19.2	20.4	22.9	22.5	21.5	18.9	15.3	12.1
		MCDB	17.0	18.1	22.0	25.5	28.6	32.8	35.0	33.5	29.8	26.9	20.9	18.3
	2%	WB	9.7	10.2	12.4	15.0	17.9	19.3	21.3	21.1	20.2	17.4	13.6	10.7
		MCDB	15.0	17.0	20.5	22.4	27.5	31.4	34.0	31.6	28.3	24.9	19.1	15.5
	5%	WB	8.6	9.0	11.5	13.9	16.8	18.5	20.4	20.2	19.2	16.5	12.4	9.7
		MCDB	13.4	15.1	19.2	21.9	26.3	30.7	33.2	31.0	27.7	23.7	18.1	14.0
	10%	WB	7.6	8.1	10.6	13.0	15.9	17.8	19.6	19.6	18.5	15.7	11.4	8.7
		MCDB	11.8	13.2	17.6	20.5	25.2	29.7	32.5	30.5	26.9	22.5	16.7	12.6
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	13.6	13.9	15.2	16.3	17.5	16.8	14.3	13.6	12.2	11.3
		MCDBR	15.7	17.1	17.9	18.3	18.4	18.5	18.9	18.8	16.8	17.0	15.6	14.9
		MCWBR	9.7	9.5	8.9	8.0	7.0	5.8	6.2	6.2	6.0	6.8	8.4	9.2
	5% WB	MCDDBR	14.4	15.3	16.3	15.8	16.8	17.4	17.3	16.8	14.8	14.0	13.8	13.3
		MCWBR	9.3	9.0	8.6	7.8	7.1	6.1	6.4	6.3	5.9	6.4	8.0	8.6
Clear-Sky Solar Irradiance	taub	0.289	0.311	0.371	0.406	0.446	0.442	0.486	0.460	0.430	0.381	0.330	0.298	
	taud	2.511	2.447	2.261	2.181	2.087	2.121	1.987	2.063	2.161	2.289	2.434	2.505	
	Ebn at Noon	919	933	894	877	844	844	803	818	822	836	856	883	
	Edh at Noon	81	97	127	146	162	157	178	162	139	112	87	77	
All-Sky Solar Radiation	RadAvg	2.64	3.46	4.67	5.74	6.62	7.43	7.51	6.57	5.16	3.98	2.87	2.36	
	RadStd	0.27	0.28	0.35	0.43	0.35	0.27	0.19	0.28	0.23	0.23	0.18	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
						+0.39	+0.39				

Nomenclature: See separate page